

Animal & tissue

KodaPure · CAT # KBTNA-A-26

IMPORTANT**NOTE BEFORE START**

1. Mix all buffers and suspensions thoroughly immediately before use. Reagents such as MB, PW, and LBIII may contain settled or undissolved material; resuspend completely to ensure consistent pipetting and reliable performance.
2. For samples with high protein content, add Proteinase K during the 56 °C incubation step, where applicable after LBII has been added. Use Proteinase K and follow the validated amount for your sample type.
3. To isolate DNA only, treat the sample with RNase A as described in the Notes section.
4. If the LBII bottle shows a substantial reduction in volume before it has been opened, do not use the reagent. Report the issue to KreatBio and request a replacement.

PROTOCOL SUMMARY

SAMPLES

**Animal tissue,
meat, organs,
cultured cells**

SAMPLE INPUT

10-20 mg tissue

HOMOGENIZATION

In LBII (see below)

ELUATE STORAGE

**4 °C short / –20 °C
long****REAGENTS SUPPLIED**

● 1 X 27 mL LBII Lysis Buffer II

● 1 X 22 mL BB Binding Buffer

● 1 X 72 mL WBIII Wash Buffer III

● 1 X 5 mL EB Elution Buffer

● 1 X 12 mL LBI Lysis Buffer I

● 1 X 0.8 mL MB Magnetic Beads

● 1 X 72 mL WBII Wash Buffer II

ADDITIONAL MATERIALS AND APPARATUS REQUIRED

- | | |
|--|--|
| <ul style="list-style-type: none">• Centrifuge machine suitable for 1.5 mL microcentrifuge tubes <hr/> | <ul style="list-style-type: none">• Pipettes and compatible pipette tips <hr/> |
| <ul style="list-style-type: none">• Vortex machine or homogenizer <hr/> | <ul style="list-style-type: none">• Proteinase K, 20 mg/mL <hr/> |
| <ul style="list-style-type: none">• Magnetic rack <hr/> | <ul style="list-style-type: none">• Heat block or incubator capable of 56 °C <hr/> |
| <ul style="list-style-type: none">• Tube rack <hr/> | <ul style="list-style-type: none">• Optional, depending on homogenization method: liquid nitrogen with chilled mortar and pestle, 2.8–4 mm steel or ceramic beads, or homogenizing stick <hr/> |
| <ul style="list-style-type: none">• 1.5 mL microcentrifuge tubes <hr/> | |

METHOD

1. Homogenize or digest the animal sample in LBII using one method below before continuing.

Cryogenic grinding

For high-quality, long-strand DNA: flash-freeze 10–20 mg sample in liquid nitrogen and grind to a fine powder using a chilled mortar and pestle. Transfer the powder immediately to a 1.5 mL tube containing 500 µL LBII. Mix well, add 5–10 µL Proteinase K, and incubate at 56 °C for 30 min.

High-speed bead shaking

For high-throughput processing: add 10–20 mg sample and 2.8–4 mm steel or ceramic beads to a tube containing 500 µL LBII. Vortex at high speed for 3 min until the tissue forms a uniform slurry. Add 5–10 µL Proteinase K and incubate at 56 °C for 30 min.

Proteinase K digestion only

For small tissue amounts: cut 10–15 mg sample into very small pieces. Add 500 µL LBII and 5–10 µL Proteinase K. Incubate at 56 °C for 1–3 h, or until the tissue dissolves. Larger samples may require overnight incubation.

Manual mincing or homogenizing stick

For pretreatment before lysis: add 10–20 mg sample and 500 µL LBII. Grind the sample to a paste-like consistency using a homogenizing stick. Add 5–10 µL Proteinase K and incubate at 56 °C or room temperature for 1–3 h.

2. Centrifuge the tube for 2 min at max speed ($\sim 12,000 \times g$). Then, transfer 200 µL supernatant into a new tube. Avoid the fat layer, pellet, and debris.
3. Add 200 µL LBI, close the tube lid and mix well. Leave it for 5 min.
4. Bring MB to room temperature and mix it well before use. Add 400 µL BB and 15 µL MB to the sample tube. Mix it and leave for 10 min.
5. Place the tube on the magnetic rack until the supernatant is clear (~ 2 min with KreatBio rack). Carefully decant or pipette out the clear supernatant without disturbing the black beads. Leave a small amount if necessary to reduce bead loss, then remove the tube from the rack.
6. Add 700 µL WBIII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant while still on the magnetic rack. Once done, remove tube from the rack. Repeat this wash if the supernatant is very dirty.
7. Add 700 µL WBII to the tube and mix well. Place it back on the magnetic rack for 1 min or until supernatant is clear. Discard the supernatant, remove the tube from the rack, and repeat this step if the supernatant is dirty (Step 7). Tip: A quick spin, then back on the rack, removes excess WBII if there is. Remove the tube from the rack.
8. Open the tube lid and leave it for 5 min to air-dry.
9. Add 50–80 µL EB to the black beads in the tube. Mix thoroughly and leave for 10 min. If some beads remain stuck to the tube wall, mix as well as possible and proceed.

-
10. The DNA/RNA is contained in the eluate. Place the tube on the magnetic rack for approximately 1 min, then transfer only the clear supernatant to a new tube leaving the black beads.
-
11. Salt or inhibitor carryover can disrupt PCR. For species ID, if samples fail to amplify, dilute 1:10 or 1:50 and rerun.



NOTES

- DNA-only extraction: (Add RNase A during or after extraction to remove RNA from supernatant):
 - **During extraction:** after Step 2, add 1 μ L of 20 mg/mL RNase A to the 200 μ L lysate and incubate at 37 °C or room temperature for 30 min. Continue with Step 3.
 - **After extraction:** add 1 μ L of 20 mg/mL RNase A to the eluate, mix, and leave at room temperature for 15 min. Your DNA is ready to use.
- Proteinase K (20 mg/mL) is required for animal tissue extraction but is not supplied.
- Choose the homogenization method according to sample toughness, throughput, and downstream DNA-length requirements. Use less starting material if the lysate is thick or difficult to clarify.

TROUBLESHOOTING

EVENT	LIKELY CAUSE	SUGGESTED FIX
Low DNA yield	Too much sample, or poor homogenization	Use less starting material and homogenise fully.
RNA in a DNA-only eluate	RNase A not used	Add RNase A and incubate as noted.
Bead carryover	Beads disturbed during transfer	Leave on the magnet until clear, then transfer carefully.
Brown eluate	Soil, fat, or debris carried over	Repeat extraction and transfer only clear lysate, avoid the pellet and fat layer. Centrifuge if it helps with separation. If this is for PCR amplification, dilute eluate to 1:10 to 1:50.
Poor PCR result	Inhibitor carryover or too much template	Dilute the eluate 1:10 to 1:50 and avoid debris.
Soil inhibition	Humic substances carried over	Use Tube A for inhibitor removal, or dilute the eluate to 1:10 to 1:50 and rerun PCR amplification.

SAFETY

SAFETY & HANDLING

- Research use only. Wear gloves, a lab coat, and eye protection.
- BB, WBII, and WBIII are flammable — keep away from heat and flame.
- LBI and WBIII contain guanidine (GuHCl) — do not mix with bleach.
- Dispose of waste according to local rules.
- TNA means total nucleic acid — these kits recover both DNA and RNA.